Chapter 6 — REST Clients and HTTP Interfaces

Chapter 5 built an API. Real services also **call** APIs: a payment provider, another team's microservice (a small, independently deployed service that owns one business area), a search engine. Calling over the network brings problems you never have with a local method call — slowness, timeouts (giving up after a set wait), partial failures (some calls or servers fail while others work), error bodies, versioning, retries. This chapter covers Spring's HTTP clients and how to use them safely:

What you'll learn

1. The four Spring HTTP clients — `RestClient`, `RestTemplate`, `WebClient`, HTTP interfaces — and which to choose (including `RestTemplate`'s deprecation timeline).
2. `RestClient` in depth: requests, bodies, `retrieve()` vs `exchange()`, `ResponseEntity`, generics.
3. Error handling: Spring's exception hierarchy, status handlers, reading Problem Details from error bodies, mapping to your own exceptions.
4. Timeouts and the underlying HTTP library (the code that actually opens connections and sends bytes: the JDK's built-in client, Apache HttpClient, Jetty, Reactor Netty) — Boot 4's `spring.http.clients.*`.
5. Interceptors (code that wraps every outgoing request) for cross-cutting concerns — things every call needs, such as headers, logging, tracing (following one request across services).
6. **HTTP service interfaces** (`@HttpExchange`) and **Boot 4's HTTP service client auto-configuration** (`@ImportHttpServices`, `spring.http.serviceclient.*`).
7. Client-side **API versioning** (Framework 7).
8. **Retries** with Framework 7 `@Retryable` — and when not to retry.
9. Testing clients: `@RestClientTest` + `MockRestServiceServer`, and tests against a real local server.

Code: `code/rest-clients/` — Spring Boot 4.1.1, Framework 7.0, Java 25. To make the tests hit a *real* HTTP server without the internet, the project also contains a small "remote" catalog API (`CatalogServerController`) that the test starts on a free local port; imagine it belongs to another team. All output shown is real.

```
<!-- RestClient.Builder, RestTemplateBuilder, HTTP service client auto-configuration (Boot 4 starter) -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-restclient</artifactId>
</dependency>
```

(An app that already has `spring-boot-starter-webmvc` for its own endpoints still adds `spring-boot-starter-restclient` for the client side — Boot 4's starters are modular.)

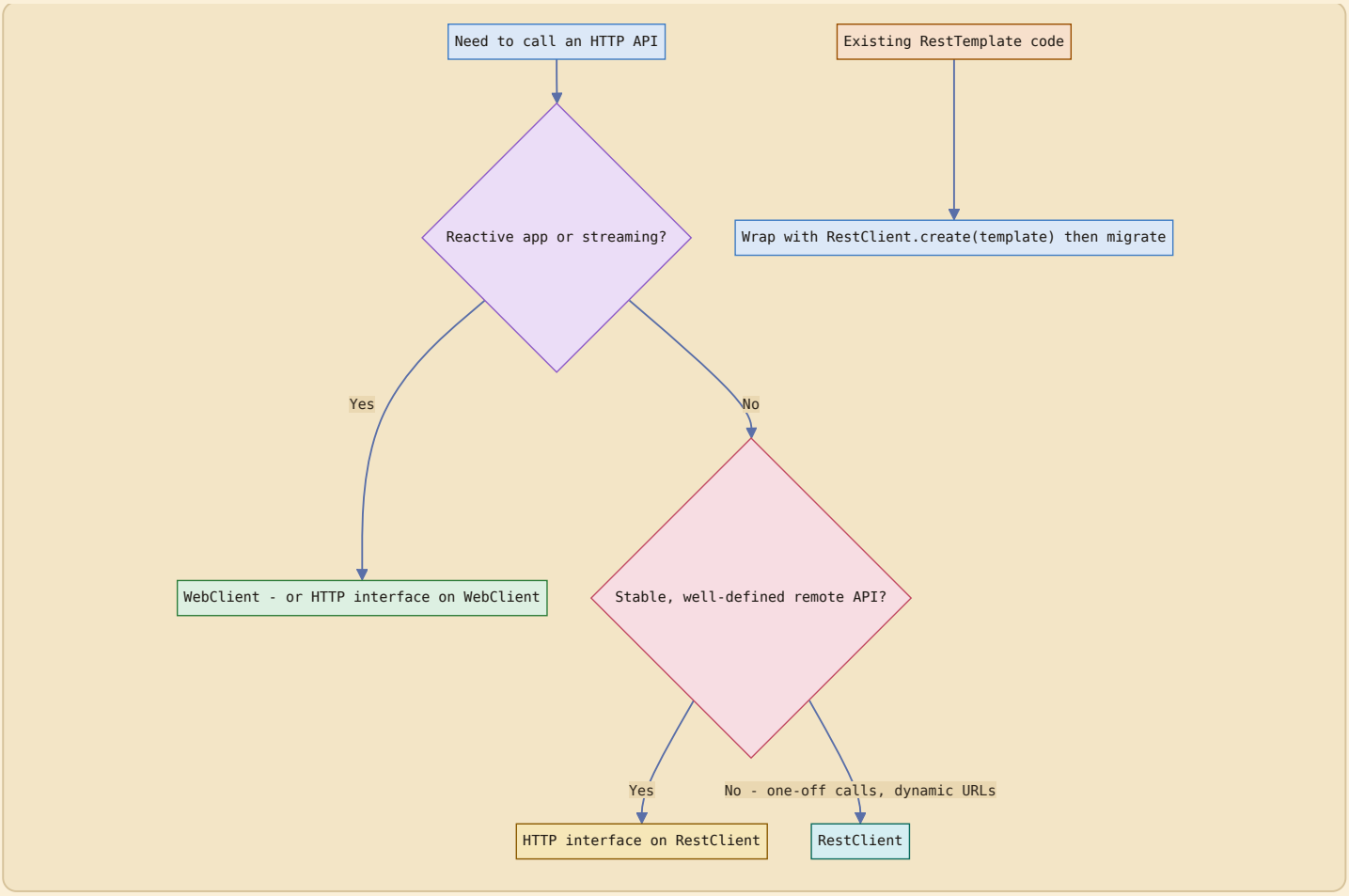
6.1 Choosing a client

What they are

Client	Style	Since	Status (Framework 7)	Use for
`RestClient`	synchronous (the calling thread waits for the response), **fluent** (calls chained with dots: `get().uri(...).retrieve()`) API	Framework 6.1 (2023)	the recommended blocking client	new code in Spring MVC / non-reactive apps
`RestTemplate`	synchronous, template methods (`getForObject`, `exchange`, ...)	Spring 3.0 (2009)	**deprecation announced**: intent to deprecate in 7.0, planned `@Deprecated` in 7.1 (Nov 2026), removal in 8.0	existing code only; migrate to `RestClient`
`WebClient`	non-blocking, reactive (the thread is freed while waiting; results arrive later as `Mono` = 0–1 value or `Flux` = 0–N values)	Framework 5.0	recommended reactive client	WebFlux apps, streaming, reactive pipelines (Chapter 13)
HTTP interface (`@HttpExchange`)	**declarative** (you describe *what* to call, not *how*): an annotated Java interface, Spring generates the implementation	Framework 6.0; Boot auto-config in 4.0	recommended for calling well-defined APIs	typed clients for other services; runs on `RestClient` or `WebClient`

Why `RestClient` replaced `RestTemplate`: the same underlying machinery (request factories, message converters, interceptors) but a modern fluent API instead of dozens of overloaded methods (methods with the same name but different parameter lists), and one place for all new features (API versioning, `requiredBody`, status handlers). The Spring team's plan (spring.io blog, "The state of HTTP clients in Spring", September 2025) keeps `RestTemplate` working through the 7.x line — so no panic, but no new code on it.

Do you need `WebClient` in a Spring MVC app? Not any more. Before `RestClient`, people pulled in WebFlux just to use `WebClient` and called `.block()`. With `RestClient` — and **virtual threads** (lightweight threads managed by the JVM, so millions can wait at once) making blocking calls cheap (Java book 15.4) — the blocking client is the simpler choice for MVC apps.



Self-check

1. Which client should new blocking code use? → `RestClient` (directly or behind an HTTP interface).
2. When is `RestTemplate` scheduled to be removed? → Framework 8.0; formally deprecated from 7.1.
3. Do you need WebFlux to use HTTP interfaces? → No; they run on `RestClient` too.

6.2 `RestClient` basics

How it works

1. Get a **builder**. In Boot, inject `RestClient.Builder` — a **prototype** bean (a new instance for every injection point) pre-configured with Boot's message converters (Jackson 3, which turns Java objects into JSON and back), the chosen HTTP library, timeouts from `spring.http.clients.*`, and (when Micrometer, Spring's metrics library, is present) observability (metrics and traces for each call). Each injection point gets a fresh builder, so configuring one does not affect others. (Outside Boot: `RestClient.builder()` or `RestClient.create("https://...")`.)
2. Set shared settings (`baseUrl`, `defaultHeader`, `defaultStatusHandler`, `requestInterceptor`, `requestFactory`) and `build()`. A `RestClient` is **immutable** (cannot change after it is built) **and thread-safe** (many threads may use it at the same time) — build it once (e.g. in a constructor) and reuse it.
3. Per call: choose the method (`get()`, `post()`, `put()`, `patch()`, `delete()`, `method(...)`), the `uri(...)`, headers, body.
4. Finish with **`retrieve()`** (then `body(...)`, `toEntity(...)`, `toBodilessEntity()`) or **`exchange(...)`**.

Example: body, entity and generic lists

```
RestClient rest = builder.baseUrl(baseUrl()).build();

BookDto book = rest.get().uri("/catalog/books/{id}", 1).retrieve().body(BookDto.class);

ResponseEntity<List<BookDto>> entity = rest.get()
        .uri(b -> b.path("/catalog/books").queryParam("author", "Martin Fowler").build())
        .retrieve()
        .toEntity(new ParameterizedTypeReference<List<BookDto>>() {});
```

Output:

```
body(BookDto.class) -> BookDto[id=1, title=Release It!, author=Michael T. Nygard]
toEntity(List) -> 200 OK 2 books
```

What each part does:

- `uri("/catalog/books/{id}", 1)` — a **URI template**; variables are URL-encoded (special characters like spaces turned into `%20`-style codes) for you. Never build URLs by string concatenation with user input.
- `uri(b -> b.path(...).queryParam("author", "Martin Fowler").build())` — the `UriBuilder` form for query parameters (the space is encoded).
- `body(BookDto.class)` — deserialises (turns the JSON text back into an object) the JSON body into your record. Your DTO (Data Transfer Object: a plain class/record that just carries data) only needs the fields you use; unknown fields are ignored (Jackson 3 default).
- `ParameterizedTypeReference<List<BookDto>>` — Java erases generics (type parameters such as `<BookDto>`) at runtime, so `List.class` would give you a list of maps; the anonymous subclass (the `{}` at the end creates a one-off subclass) captures the full type.
- `toEntity(...)` returns `ResponseEntity<T>`: status + headers + body. `toBodilessEntity()` when you only care about the status.
- `requiredBody(...)` (Framework 7.0.4+) is like `body(...)` but throws if the body is empty instead of returning `null`.

A hand-written client class

Wrap the calls in a small class so the rest of your code talks in domain terms (the language of the business — books, orders — rather than HTTP) (`CatalogClient.java`):

```java
@Component
public class CatalogClient {

    private final RestClient rest;

    public CatalogClient(RestClient.Builder builder, @Value("${catalog.base-url}") String baseUrl) {
        this.rest = builder                                    // Boot's pre-configured, prototype builder
                .baseUrl(baseUrl)
                .defaultHeader("X-Client", "catalog-client")
                .defaultStatusHandler(status -> status.is5xxServerError(), (request, response) -> {
                    throw new CatalogUnavailableException(
                            "catalog answered " + response.getStatusCode().value() + " for " +
request.getURI().getPath());
                })
                .build();
    }

    /** 404 becomes Optional.empty(); 5xx becomes CatalogUnavailableException (default status handler). */
    public Optional<BookDto> findBook(long id) {
        try {
            return Optional.ofNullable(rest.get()
                    .uri("/catalog/books/{id}", id)           // URI template: id is encoded for us
                    .retrieve()                               // status handlers apply to retrieve()
                    .body(BookDto.class));
        } catch (HttpClientErrorException.NotFound e) {
            return Optional.empty();
        }
    }

    /** exchange(): full control over the raw response - status handlers are NOT applied. */
    public String describe(long id) {
        return rest.get()
                .uri("/catalog/books/{id}", id)
                .exchange((request, response) -> {
                    if (response.getStatusCode().isError()) {
                        return "status " + response.getStatusCode().value();
                    }
                    BookDto b = response.bodyTo(BookDto.class);
                    return b.title() + " by " + b.author();
                });
    }

    public ResponseEntity<BookDto> create(String title, String author) {
        return rest.post()
                .uri("/catalog/books")
                .body(new BookDto(null, title, author))      // serialised to JSON by Jackson
                .retrieve()
                .toEntity(BookDto.class);                     // status + headers + body
    }
}
```

Output:

```
findBook(2)   -> Optional[BookDto[id=2, title=Refactoring, author=Martin Fowler]]
findBook(999) -> Optional.empty
describe(1)   -> Release It! by Michael T. Nygard
describe(999) -> status 404
create        -> 201 CREATED Location=/catalog/books/102 body=BookDto[id=102, title=Designing Data-Intensive
Applications, author=Martin Kleppmann]
```

`retrieve()` vs `exchange()`

	`retrieve()`	`exchange((req, res) -> ...)`
Error statuses	turned into exceptions (default or your status handlers)	**you** inspect the status; **status handlers are not applied**
Body handling	`body`, `toEntity`, `toBodilessEntity`	`res.bodyTo(...)`, `res.getHeaders()`, anything
Use when	almost always	different body types per status, streaming, very custom handling

A lesson from writing this chapter: the first version of `findBook` used `exchange()` and treated 404 specially, assuming the 5xx (HTTP status codes 500–599: server-side failures) `defaultStatusHandler` would still fire. It does not — `exchange()` bypasses status handlers — so a 503 silently became `Optional.empty()`. The `@RestClientTest` in section 6.8 caught it. If you use `exchange()`, handle **every** status yourself.

Self-check

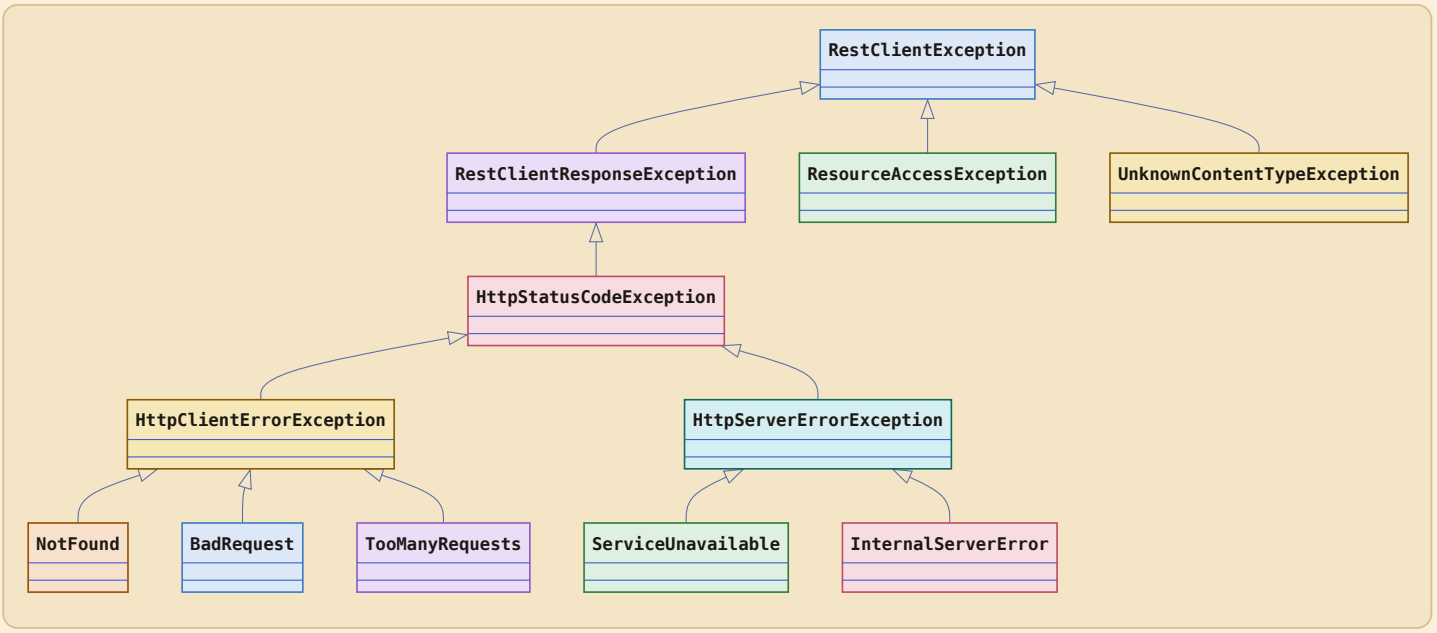
1. Why inject `RestClient.Builder` instead of calling `RestClient.builder()` in Boot? → Boot's builder carries the auto-configured converters, HTTP library, timeouts and observability.
2. Is a `RestClient` safe to share between threads? → Yes, it is immutable and thread-safe once built.
3. How do you deserialise a `List<BookDto>`? → `body(new ParameterizedTypeReference<List<BookDto>>() {})`.

More request features (quick reference)

Need	`RestClient` call
Header on one request	`.header("X-Tenant", "acme")`, `.headers(h -> h.setBearerAuth(token))`
Accept / content type	`.accept(MediaType.APPLICATION_JSON)`, `.contentType(MediaType.APPLICATION_JSON)`
Form post (`application/x-www-form-urlencoded`)	`.contentType(APPLICATION_FORM_URLENCODED).body(multiValueMap)`
Multipart upload	`.body(new MultipartBodyBuilder().part("file", resource).build())` with `contentType(MULTIPART_FORM_DATA)`
Download a large file	`.exchange((req, res) -> { res.getBody().transferTo(out); return null; })` — stream, don't buffer
Plain text	`.retrieve().body(String.class)`
Per-request attributes for interceptors/observations	`.attribute("name", value)`

6.3 Error handling

Spring's client exception hierarchy



Exception	Meaning	Retry?
`HttpClientErrorException` (4xx) with nested types `.NotFound`, `.BadRequest`, `.Conflict`, `.TooManyRequests`, ...	the **request** was wrong or not allowed	no (except 429 "too many requests" after waiting the `Retry-After` header's delay, and 408 "request timeout")
`HttpServerErrorException` (5xx) with `.InternalServerError`, `.BadGateway`, `.ServiceUnavailable`, `.GatewayTimeout`	the server failed	often yes (transient — likely to pass on its own), if the operation is idempotent (safe to repeat: doing it twice has the same effect as once — see System Design Fundamentals)
`ResourceAccessException`	I/O (input/output, here network) problem: connection refused, **timeout**, DNS (the system that turns host names into IP addresses)	often yes
`UnknownContentTypeException`	no converter for the response type	no — a bug or contract change (the other side changed its API)

All are **unchecked** (you are not forced to catch or declare them: `RestClientException extends RuntimeException`). `RestClientResponseException` gives you `getStatusCode()`, `getResponseHeaders()`, `getResponseBodyAsString()`, and **`getResponseBodyAs(Class)`** — which can decode an RFC 9457 error body directly:

```
assertThatThrownBy(() -> rest.get().uri("/catalog/books/{id}", 999).retrieve().body(BookDto.class))
        .isInstanceOf(HttpClientErrorException.NotFound.class)
        .satisfies(e -> {
            var ex = (HttpClientErrorException) e;
            ProblemDetail pd = ex.getResponseBodyAs(ProblemDetail.class);   // read the error body
            System.out.println("404 -> " + ex.getClass().getSimpleName()
                    + ", problem title='" + pd.getTitle() + "', detail='" + pd.getDetail() + "'");
        });
```

Output:

```
404 -> NotFound, problem title='Book not found', detail='No book 999'
```

Mapping to your own exceptions

Callers of `CatalogClient` should not need to know it uses HTTP. Translate at the edge:

- **Per request:** `.retrieve().onStatus(HttpStatusCode::is4xxClientError, (req, res) -> { throw new ... })`.
- **For every request of a client:** `builder.defaultStatusHandler(predicate, handler)` — used above to turn any 5xx into `CatalogUnavailableException`:

```
503 -> CatalogUnavailableException: catalog answered 503 for /catalog/flaky
```

- **Expected "not found":** return `Optional.empty()` (as `findBook` does) instead of throwing.

Common mistakes

- Catching `Exception` around a client call and returning `null` — hides outages; log and translate instead.
- Logging full response bodies of errors — may contain personal data or secrets.
- Treating every 4xx as retryable — a 400 will fail the same way every time.
- Forgetting that **no timeout = wait forever**-ish (next section).

Self-check

1. A call times out. Which exception? → `ResourceAccessException` (its cause is the library's timeout exception, e.g. `HttpTimeoutException` for the JDK client).
2. How do you read an RFC 9457 error body from a failed call? → `ex.getResponseBodyAs(ProblemDetail.class)` on the `RestClientResponseException`.
3. Where should HTTP exceptions be translated into domain exceptions? → At the client edge (status handlers or the client class), so business code never sees HTTP types.

6.4 Timeouts and the underlying HTTP library

What it is

`RestClient` does not open sockets (an operating-system network connection endpoint) itself; it delegates to a `ClientHttpRequestFactory` wrapping a real HTTP library. Two timeouts matter:

- **Connect timeout** — how long to wait for the TCP (the protocol that sets up a reliable connection) and TLS (the encryption layer behind `https`) connection to be established.
- **Read timeout** — how long to wait for response data once connected.

Why it matters: without timeouts, a slow downstream service (one that *you* call) makes your request threads pile up waiting, until your service also stops answering — a **cascading failure** (one failure spreading to the services that depend on it). Every remote call needs a timeout shorter than your own callers' timeout. See also Circuit breaker and Bulkhead.

Which library does Boot pick?

Boot 4 detects, in this order, the first one on the classpath:

Priority	Library	Factory	Add dependency
1	Apache HttpClient 5	`HttpComponentsClientHttpRequestFactory`	`org.apache.httpcomponents.client5:httpclient5`
2	Jetty HttpClient	`JettyClientHttpRequestFactory`	`org.eclipse.jetty:jetty-client`
3	Reactor Netty	`ReactorClientHttpRequestFactory`	`io.projectreactor.netty:reactor-netty-http`
4	JDK `java.net.http.HttpClient`	`JdkClientHttpRequestFactory`	nothing (built in)
5	`HttpURLConnection`	`SimpleClientHttpRequestFactory`	nothing

This project has none of the first three, so it uses the **JDK HttpClient**. Force a choice with `spring.http.clients.imperative.factory=jdk|http-components|jetty|reactor|simple`. (For `WebClient` the equivalent is `spring.http.clients.reactive.connector`.)

Configuring timeouts

Globally, for every auto-configured client (Boot 4 property names — the Boot 3.x `spring.http.client.*` names are deprecated):

```
spring.http.clients.connect-timeout=1s
spring.http.clients.read-timeout=3s
spring.http.clients.redirects=dont-follow     # or follow / follow-when-possible
```

Per client, in code, with Boot's `ClientHttpRequestFactoryBuilder` and `HttpClientSettings`:

```
var factory = ClientHttpRequestFactoryBuilder.jdk()
        .build(HttpClientSettings.defaults()
                .withConnectTimeout(Duration.ofSeconds(1))
                .withReadTimeout(Duration.ofMillis(300)));
RestClient rest = builder.baseUrl(baseUrl()).requestFactory(factory).build();
rest.get().uri("/catalog/slow").retrieve().body(String.class);   // server needs 1.5 s
```

Output:

```
slow endpoint -> ResourceAccessException caused by HttpTimeoutException after ~300 ms
```

For HTTP service groups (section 6.6) the timeouts can be set per group: `spring.http.serviceclient.<group>.read-timeout=2s`.

Choosing values: measure the downstream's normal latency (how long one call takes; p99 = the time that 99% of calls stay under) and set the read timeout a bit above it; keep the total (timeout × attempts + back-off — the waits between retries) below what *your* caller will wait. Connection **pooling** (keeping open connections and reusing them) (Apache/Jetty/Netty pool by default; the JDK client keeps connections alive) avoids paying TCP/TLS setup on every call.

Self-check

1. Connect vs read timeout? → Connect: establishing the connection. Read: waiting for response data after connecting.
2. Which HTTP library does Boot use if you add nothing? → The JDK `java.net.http.HttpClient`.
3. Boot 4 property for a global read timeout? → `spring.http.clients.read-timeout` (the Boot 3 name `spring.http.client.read-timeout` is deprecated).

6.5 Interceptors

A `ClientHttpRequestInterceptor` wraps every request a client sends — the client-side twin of a servlet filter. Uses: auth headers (tokens — strings proving who the caller is), correlation ids (one id carried through all calls of a request so logs can be joined), logging, metrics.

```
RestClient rest = builder.baseUrl(baseUrl())
        .requestInterceptor((request, body, execution) -> {
            request.getHeaders().add("X-Trace-Id", "trace-42");          // add a header
            var response = execution.execute(request, body);            // do the call
            log.add(request.getMethod() + " " + request.getURI().getPath() + " -> " + response.getStatusCode().value());
            return response;
        })
        .build();
```

Output:

```
interceptor: GET /catalog/books/1 -> 200
interceptor: GET /catalog/books/3 -> 200
```

- You rarely need interceptors for **tracing**: when Micrometer observability is set up (Actuator, Chapter 15), Boot instruments the auto-configured builder with an observation customizer that records `http.client.requests` metrics and propagates trace headers (passes the current trace id on to the next service). This only works if you build from the injected builder — `RestClient.create()` skips it.

- For OAuth2 bearer tokens (access tokens sent as `Authorization: Bearer ...`; OAuth2 is the standard protocol for issuing them), Spring Security provides `OAuth2ClientHttpRequestInterceptor` (Chapter 11).
- Don't read the response body in an interceptor unless you buffer it (`BufferingClientHttpRequestFactory`); a body can normally be read only once.
- To customise **all** builders Boot creates, declare a `RestClientCustomizer` bean.

6.6 HTTP service interfaces (`@HttpExchange`)

What it is

An **HTTP service interface** is a Java interface whose methods are annotated with the HTTP method, path and parameters. Spring generates a **proxy** that implements it by turning each call into an HTTP request (the same proxy idea as Chapter 4). It is Spring's built-in answer to Feign-style declarative clients (Feign/OpenFeign is a popular library that does the same from annotated interfaces) — no extra library. Analogy: a TV remote. You press "channel 5"; you do not care which infrared codes are sent.

```
@HttpExchange("/catalog")
public interface CatalogApi {

    @GetExchange("/books/{id}")
    BookDto get(@PathVariable long id);

    @GetExchange("/books/{id}")
    ResponseEntity<BookDto> getEntity(@PathVariable long id);   // status + headers too

    @GetExchange("/books")
    List<BookDto> byAuthor(@RequestParam String author);

    @PostExchange("/books")
    ResponseEntity<BookDto> create(@RequestBody BookDto book);

    @GetExchange(url = "/greeting", version = "2.0")   // Framework 7: API version per method
    String greetingV2();
}
```

Annotation	Purpose
`@HttpExchange` (type)	shared URL prefix, content type, accept, headers, version
`@GetExchange`, `@PostExchange`, `@PutExchange`, `@PatchExchange`, `@DeleteExchange`	method + path
`@PathVariable`, `@RequestParam`, `@RequestHeader`, `@RequestBody`, `@RequestPart`, `@CookieValue`	the same annotations as server controllers, now describing the request to send
`URI` parameter	override the full URL dynamically

Return types: a body type (`BookDto`, `List<BookDto>`), `ResponseEntity<T>`, `ResponseEntity<Void>`, `HttpHeaders`, `void`; on `WebClient`, also `Mono<T>` / `Flux<T>`. Errors surface as the same `HttpClientErrorException` / `HttpServerErrorException` as `RestClient`.

Boot 4 auto-configuration: `@ImportHttpServices`

Before Framework 7 / Boot 4 you created each proxy yourself. Now you declare **groups** of interfaces; each group gets its own `RestClient` configured from properties:

```
@SpringBootApplication
@EnableResilientMethods                                  // for @Retryable (Framework 7)
@ImportHttpServices(group = "catalog", types = CatalogApi.class)   // creates a CatalogApi bean
public class ClientsApplication { ... }
```

```
# Boot 4: HTTP service client group "catalog" (the @HttpExchange interface)
spring.http.serviceclient.catalog.base-url=http://localhost:8080
spring.http.serviceclient.catalog.read-timeout=2s
spring.http.serviceclient.catalog.default-header.X-Client=handbook-ch06
# the group's RestClient sends the API version (from @GetExchange(version=...)) in this header
spring.http.serviceclient.catalog.apiversion.insert.header=X-API-Version
```

- `types = ...` lists interfaces; `basePackages` / `basePackageClasses` scans a package instead. Repeat the annotation for several groups (one per remote service).
- `clientType = WEB_CLIENT` makes a group use `WebClient`.
- For configuration beyond properties, declare a `RestClientHttpServiceGroupConfigurer` bean: `groups -> groups.filterByName("catalog").forEachClient((group, builder) -> builder.requestInterceptor(...))`.

Then inject and call it like any bean (`HttpInterfaceTest.java`):

```
@Autowired CatalogApi catalogApi;     // generated by @ImportHttpServices + Boot auto-config

BookDto one = catalogApi.get(1);
ResponseEntity<BookDto> e = catalogApi.getEntity(2);
ResponseEntity<BookDto> created = catalogApi.create(new BookDto(null, "Accelerate", "Nicole Forsgren"));
String v2 = catalogApi.greetingV2();
```

Output:

```
bean class     -> $ProxyN
get(1)         -> BookDto[id=1, title=Release It!, author=Michael T. Nygard]
byAuthor(...)  -> [BookDto[id=2, title=Refactoring, author=Martin Fowler], BookDto[id=3, title=Patterns of Enterprise
Application Architecture, author=Martin Fowler]]
getEntity(2)   -> 200 OK server saw X-Client=handbook-ch06
create(...)    -> 201 CREATED BookDto[id=101, title=Accelerate, author=Nicole Forsgren]
greetingV2()   -> v2: hello
get(999)       -> NotFound
```

The bean is a JDK dynamic proxy (`$Proxy...`). `X-Client=handbook-ch06` shows the per-group `default-header` property reached the server, and `greetingV2()` shows the method's `version = "2.0"` was sent in the `X-API-Version` header configured for the group.

What Boot does for you, by hand

```
RestClient restClient = RestClient.builder().baseUrl(baseUrl()).build();
HttpServiceProxyFactory factory = HttpServiceProxyFactory
        .builderFor(RestClientAdapter.create(restClient)).build();
CatalogApi manual = factory.createClient(CatalogApi.class);
```

```
manual proxy get(3) -> Patterns of Enterprise Application Architecture
```

Useful in plain Spring, in tests, or in libraries.

Trade-offs

	HTTP interface	Hand-written `RestClient` class	OpenFeign (Spring Cloud)
Boilerplate	minimal	more	minimal
Custom per-call logic (fallbacks, `Optional` on 404)	needs a wrapper or default methods (interface methods with a body)	natural	via fallbacks (alternative answers returned when the call fails)
Dependencies	none beyond Spring	none	Spring Cloud OpenFeign (in maintenance mode — bug fixes only, no new features; Spring recommends HTTP interfaces)
Config per remote service	`spring.http.serviceclient.<group>.*` (Boot 4)	your own properties	`spring.cloud.openfeign.client.config.*`
Can share the interface with the server?	technically, but couples client and server release cycles (neither can ship a change without the other)	—	same caveat

Common approach: an HTTP interface for the raw calls + a thin service class on top for domain translation (`Optional`, own exceptions, retries).

Self-check

1. What does `@ImportHttpServices(group = "catalog", types = CatalogApi.class)` do? → Registers a `CatalogApi` proxy bean backed by a `RestClient` for group `catalog`, configured from `spring.http.serviceclient.catalog.*`.
2. Which class creates the proxy under the hood? → `HttpServiceProxyFactory` with a `RestClientAdapter` (or `WebClientAdapter`).
3. How would you add an interceptor to only one group? → A `RestClientHttpServiceGroupConfigurer` bean using `groups.filterByName("catalog").forEachClient(...)`.

6.7 Client-side API versioning and retries

API versioning (Framework 7)

When the server versions its API (Chapter 5, section 5.9), the client must send the version. Framework 7 adds this to every client: an `ApiVersionInserter` decides *how* (header, query param, path segment, media-type parameter), `defaultApiVersion` sets the usual version, and each request can override it with `.apiVersion(...)`:

```
RestClient rest = builder.baseUrl(baseUrl())
        .apiVersionInserter(ApiVersionInserter.useHeader("X-API-Version"))  // how to send it
        .defaultApiVersion("1.0")                                           // unless overridden
        .build();
String v1 = rest.get().uri("/catalog/greeting").retrieve().body(String.class);
String v2 = rest.get().uri("/catalog/greeting").apiVersion("2.0").retrieve().body(String.class);
```

Output:

```
default version -> v1: hello
apiVersion(2.0) -> v2: hello
```

HTTP interfaces put the version on the method (`@GetExchange(version = "2.0")`) and configure the inserter per group (`spring.http.serviceclient.<group>.apiversion.insert.header=...`), as shown in 6.6. Server-side version properties (`spring.mvc.apiversion.*`) are **not** applied to clients; configure them separately.

Retries

What it is

A **retry** repeats a failed call hoping the failure was transient. Framework 7's `@Retryable` (Chapter 4, section 4.7) makes this one annotation:

```
@Service
public class StockService {
    private final RestClient rest;

    public StockService(RestClient.Builder builder, @Value("${catalog.base-url}") String baseUrl) {
        this.rest = builder.baseUrl(baseUrl).build();
    }

    @Retryable(includes = org.springframework.web.client.HttpServerErrorException.class,
               maxRetries = 3, delay = 20, multiplier = 2, jitter = 5)
    public String checkStock() {
        return rest.get().uri("/catalog/flaky").retrieve().body(String.class);
    }
}
```

The `/catalog/flaky` endpoint answers 503 twice, then 200:

```
checkStock() -> ok on call 3
```

Rules for safe retries

1. **Only transient failures:** 5xx (especially 502/503/504), timeouts/`ResourceAccessException`, 429 (respect `Retry-After`). Never 400/401/403/404/409/422.
2. **Only idempotent operations** — GET, PUT, DELETE — or POSTs protected by an **idempotency key** header that the server uses to de-duplicate (the way payment APIs such as Stripe do).
3. **Exponential back-off + jitter** (waits that grow each time, plus a small random amount) so thousands of clients don't hammer a recovering service in sync.
4. **Bound the total time**: attempts × read timeout + delays must fit your own deadline (`@Retryable(timeout = ...)` since 7.0.2 caps the total).
5. **Retry in one layer only**, and combine with a **circuit breaker** (a switch that stops calling a failing service for a while; Resilience4j is a common library for it) so a dead dependency is not retried forever — see Chapter 16 and Retry pattern.
6. `@Retryable` is proxy-based: it must be on a Spring bean method called from **another** bean.

Self-check

1. Should you retry a `POST /payments` after a read timeout? → Only if the request carries an idempotency key the server honours; otherwise you may charge twice.
2. With `maxRetries = 3`, how many HTTP calls at most? → 4.
3. Does the server's `spring.mvc.apiversion.use.header` configure your clients? → No; configure `apiVersionInserter` or the service-group `apiversion.*` properties.

6.8 Testing clients

Option 1 — `@RestClientTest` + `MockRestServiceServer` (fast, no network)

`@RestClientTest` is a **test slice** (a test that starts only a small part of the application): it starts only the listed components, Jackson and the `RestClient.Builder`/`RestTemplateBuilder` auto-configuration, and binds a `MockRestServiceServer` to the builder. Requests never leave the JVM; the mock checks each request against expectations and returns canned responses (fixed answers prepared in the test). Boot 4 dependency: `spring-boot-starter-restclient-test`.

```java
@RestClientTest(components = CatalogClient.class, properties = "catalog.base-url=http://catalog.test")
class CatalogClientMockServerTest {

    @Autowired CatalogClient client;
    @Autowired MockRestServiceServer server;   // intercepts requests; no network at all

    @Test
    void foundNotFoundAndUnavailable() {
        server.expect(requestTo("http://catalog.test/catalog/books/1"))
              .andExpect(method(HttpMethod.GET))
              .andExpect(header("X-Client", "catalog-client"))
              .andRespond(withSuccess("""
                     {"id":1,"title":"Release It!","author":"Michael T. Nygard","pages":376}""",
                     MediaType.APPLICATION_JSON));
        server.expect(requestTo("http://catalog.test/catalog/books/2"))
              .andRespond(withStatus(HttpStatus.NOT_FOUND));
        server.expect(requestTo("http://catalog.test/catalog/books/3"))
              .andRespond(withStatus(HttpStatus.SERVICE_UNAVAILABLE));

        System.out.println("mock 200 -> " + client.findBook(1));
        System.out.println("mock 404 -> " + client.findBook(2));
        assertThatThrownBy(() -> client.findBook(3))
                .isInstanceOf(CatalogUnavailableException.class)
                .satisfies(e -> System.out.println("mock 503 -> " + e.getMessage()));
        server.verify();                          // every expected request really happened
    }
}
```

Output:

```
mock 200 -> Optional[BookDto[id=1, title=Release It!, author=Michael T. Nygard]]
mock 404 -> Optional.empty
mock 503 -> catalog answered 503 for /catalog/books/3
```

The unknown `pages` field was ignored. By default expectations must occur **in order**; `MockRestServiceServer.bindTo(builder).ignoreExpectOrder(true)` relaxes that. `withException(new SocketTimeoutException())` simulates I/O failures.

Option 2 — a real HTTP server

Mocks can't catch problems in the real HTTP layer (timeouts, headers, encoding, the version header). This chapter's other tests start the app on a real port with a local "remote" controller:

```java
@SpringBootTest(webEnvironment = SpringBootTest.WebEnvironment.DEFINED_PORT)
abstract class LiveServerTest {

    static final int PORT = freePort();

    @DynamicPropertySource
    static void urls(DynamicPropertyRegistry registry) {
        String base = "http://localhost:" + PORT;
        registry.add("server.port", () -> PORT);
        registry.add("catalog.base-url", () -> base);
        registry.add("spring.http.serviceclient.catalog.base-url", () -> base);
    }
    ...
}
```

The port is picked **before** the context starts so the client base URLs can be set as properties (with `RANDOM_PORT` the port is only known after the clients were created). In real projects the "remote" side is usually a **WireMock** server (a standalone HTTP stub server, i.e. a fake server returning prepared responses; often run via Testcontainers, a library that starts Docker containers for tests) or a Spring Cloud Contract stub (a fake generated from an agreed API contract) — see Chapter 12.

For testing **your own** endpoints over HTTP, Boot 4 offers **`RestTestClient`** (`@AutoConfigureRestTestClient`), the `RestClient`-based successor of `TestRestTemplate` (which now needs `@AutoConfigureTestRestTemplate` and the `spring-boot-resttestclient` module).

Self-check

1. What does `server.verify()` check? → That every expected request was actually made.
2. Why also test against a real (stub) server? → Mocks skip the real HTTP layer: timeouts, encoding, headers, connection handling.
3. Which Boot 4 starter provides `@RestClientTest`? → `spring-boot-starter-restclient-test`.

6.9 `RestTemplate` and `WebClient` in brief

`RestTemplate` (legacy)

```
RestTemplate template = restTemplateBuilder.rootUri(baseUrl()).build();
BookDto viaTemplate = template.getForObject("/catalog/books/{id}", BookDto.class, 2);

RestClient wrapped = RestClient.create(template);     // migration step: reuse its config
BookDto viaWrapped = wrapped.get().uri("/catalog/books/{id}", 2).retrieve().body(BookDto.class);
```

```
RestTemplate  -> BookDto[id=2, title=Refactoring, author=Martin Fowler]
RestClient.create(template) -> BookDto[id=2, title=Refactoring, author=Martin Fowler]
```

Boot auto-configures a `RestTemplateBuilder` (not a `RestTemplate`). Migration map:

`RestTemplate`	`RestClient`
`getForObject(url, T.class, vars)`	`get().uri(url, vars).retrieve().body(T.class)`
`getForEntity(...)`	`get().uri(...).retrieve().toEntity(T.class)`
`postForEntity(url, body, T.class)`	`post().uri(url).body(body).retrieve().toEntity(T.class)`
`exchange(url, method, entity, T.class)`	`method(m).uri(url).headers(...).body(...).retrieve().toEntity(T.class)`
`setErrorHandler(...)`	`defaultStatusHandler(...)`
interceptors / request factory	reused as-is (`requestInterceptor`, `requestFactory`)

Steps: (1) wrap existing templates with `RestClient.create(restTemplate)` or `RestClient.builder(restTemplate)` so configuration is reused; (2) rewrite call sites; (3) move shared config to a `RestClientCustomizer` / builder; (4) remove the templates.

`WebClient` (reactive) — not run here

```
WebClient web = webClientBuilder.baseUrl("https://catalog.example.com").build();
Mono<BookDto> book = web.get().uri("/catalog/books/{id}", 1)
        .retrieve()
        .bodyToMono(BookDto.class)
        .timeout(Duration.ofSeconds(2));
```

Same shape as `RestClient`, but returns `Mono`/`Flux` and never blocks a thread. Use it in WebFlux apps or for streaming; don't call `.block()` inside a reactive pipeline. Full coverage in Chapter 13.

6.10 Common mistakes, real-world examples, interview

Common mistakes

Mistake	Consequence	Fix
New `RestClient` per request	loses connection pooling, slower, more sockets	build once, reuse
`RestClient.create()` in a Boot app	no Boot converters/timeouts/observability	inject `RestClient.Builder`
No timeouts	thread pile-up, cascading failure	`spring.http.clients.*-timeout` or per-group/per-client settings
String-concatenated URLs	broken encoding, injection (attacker-controlled text changing the meaning of the URL)	URI templates / `UriBuilder`
Using `exchange()` and expecting status handlers	errors silently treated as success	handle statuses yourself or use `retrieve()`
Retrying POSTs / 4xx	duplicate orders, wasted calls	retry only idempotent + transient
Leaking `HttpClientErrorException` into business code	HTTP details spread everywhere	translate at the client edge
Mock-only tests	real HTTP issues missed	add a test against a real/stub server

Real-world examples

- **Payment integration**: a `PaymentsApi` HTTP interface for the provider, read timeout ~5 s, retries only for GETs and for POSTs carrying an `Idempotency-Key`, errors mapped to `PaymentDeclined` vs `PaymentProviderUnavailable`.
- **Microservice calls** inside a company: one `@ImportHttpServices` group per downstream service (`inventory`, `pricing`, `shipping`) each with its own base URL and timeouts from config, tracing headers propagated automatically.
- **Third-party APIs with versions** (e.g. APIs that version by header or date): a default API version on the client, bumped deliberately after testing.

In the interview

Q: Which HTTP client would you use in a new Spring Boot 4 MVC service, and why? 30-second answer: "`RestClient` — or an `@HttpExchange` interface backed by it. It's the modern synchronous client: fluent, uses Boot's auto-configured converters, HTTP library, timeouts and observability. `RestTemplate` is being deprecated (formally in 7.1, removal in 8.0), and `WebClient` is for reactive stacks. With virtual threads, blocking calls are cheap enough that I don't need reactive just for concurrency."

- *How do HTTP interfaces get implemented?* → `HttpServiceProxyFactory` creates a JDK dynamic proxy; in Boot 4 `@ImportHttpServices` registers groups and `spring.http.serviceclient.<group>.*` configures each group's client.
- *How do you handle errors?* → `retrieve()` throws `HttpClientErrorException`/`HttpServerErrorException` (with nested per-status types) and `ResourceAccessException` for I/O/timeouts; I map them to domain exceptions with `onStatus`/`defaultStatusHandler` and read `ProblemDetail` bodies with `getResponseBodyAs`.
- *What timeouts do you set?* → Connect and read timeouts on every client (`spring.http.clients.connect-timeout/read-timeout` or per group), sized from downstream p99 and my own SLA (Service Level Agreement: the response time/availability promised to callers).

Q: How do you make remote calls resilient? → Timeouts, retries with exponential back-off + jitter only for idempotent/transient failures (`@Retryable` in Framework 7), circuit breaker + bulkhead (Resilience4j / `@ConcurrencyLimit`), fallbacks where acceptable, and idempotency keys for writes.

Q: How do you test a REST client? → `@RestClientTest` + `MockRestServiceServer` for fast request/response contract checks; plus an integration test against a real stub server (WireMock) or the provider's contract stubs for the HTTP layer.

6.11 Chapter TL;DR

- **Clients:** `RestClient` (sync, recommended) · HTTP interfaces `@HttpExchange` (declarative, on `RestClient` or `WebClient`) · `WebClient` (reactive) · `RestTemplate` (legacy; deprecated from 7.1, removed in 8.0).
- **Boot 4:** add `spring-boot-starter-restclient`; inject the **prototype** `RestClient.Builder`; build once, reuse (thread-safe).
- **Calls:** `get().uri(template, vars).retrieve().body(T)` / `toEntity(T)` / `toBodilessEntity()`; `ParameterizedTypeReference` for generics; `exchange()` for raw control (**status handlers not applied**).
- **Errors:** 4xx → `HttpClientErrorException.*`, 5xx → `HttpServerErrorException.*`, I/O/timeouts → `ResourceAccessException`; `getResponseBodyAs(ProblemDetail.class)`; translate with `onStatus`/`defaultStatusHandler`.
- **Timeouts:** `spring.http.clients.connect-timeout`/`read-timeout` (Boot 4 names), per group `spring.http.serviceclient.<g>.read-timeout`, or `ClientHttpRequestFactoryBuilder` + `HttpClientSettings`. Library order: Apache → Jetty → Reactor Netty → JDK → simple.
- **HTTP interfaces in Boot 4:** `@ImportHttpServices(group = "...", types = ...)` + `spring.http.serviceclient.<group>.base-url/default-header/apiversion.*`; customise with `RestClientHttpServiceGroupConfigurer`.
- **Versioning (Framework 7):** `ApiVersionInserter.useHeader(...)` + `defaultApiVersion(...)`, per-request `.apiVersion(...)`, `@GetExchange(version = ...)`.
- **Retries:** `@Retryable` for transient + idempotent calls only, with back-off, jitter and a total budget; add a circuit breaker.
- **Tests:** `@RestClientTest` + `MockRestServiceServer`; real/stub server tests for the HTTP layer; `RestTestClient` for your own endpoints.
- Next: Chapter 7 — Data access with JDBC and transactions.